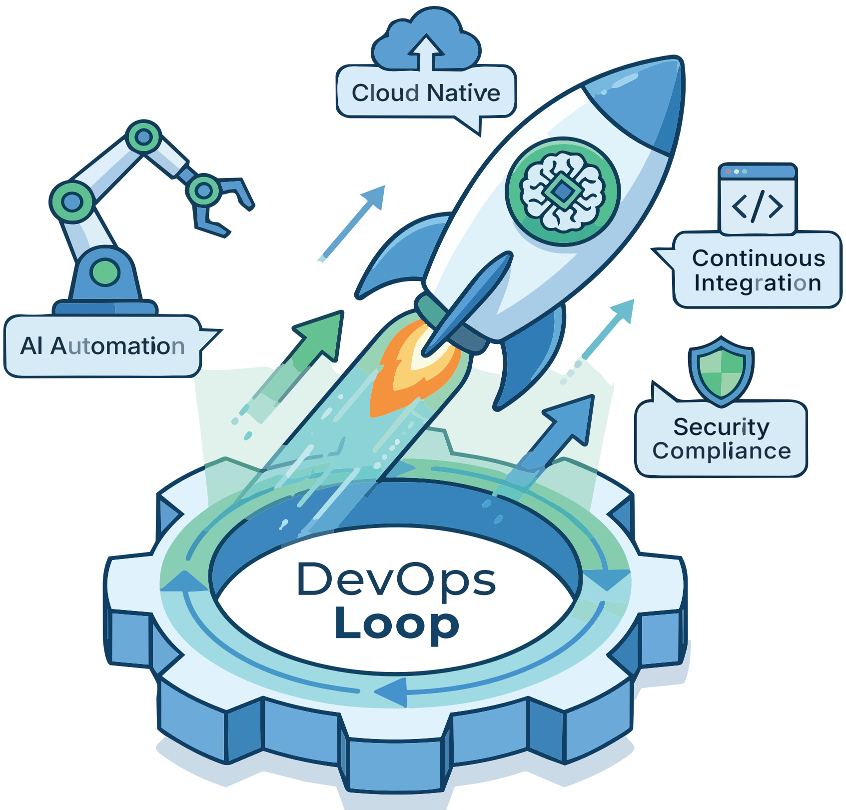
AI-based Tools: Accelerating DevOps Adoption

Modern software engineering is getting a boost with the arrival of a range of AI-based open source tools that enhance the functioning of DevOps.



DevOps is about creating a collaborative environment that improves software delivery and increases business value. Its objective is to maximise the predictability, efficiency, security and maintainability of operational processes.

DevOps targets product delivery, quality testing, feature development and maintenance releases to improve software reliability, security, and faster development and deployment cycles.

The goals of DevOps include:

- Improved deployment frequency
- Faster time to market
- Lower failure rates of new releases
- Faster recovery time from crashes or failures

The latest AI-enabled DevOps strategies are:

Using platform engineering: DevOps practices are increasingly being delivered through internal developer platforms, self-service workflows and reusable paved roads. The goal is to hide infrastructure complexity without hiding engineering accountability.

DevSecOps to secure the software supply chain: Security is moving into the delivery system through SAST/DAST, dependency scanning, SBOMs, artifact signing, provenance and Policy as Code.

Continuous delivery engineering: Organisations are optimising the entire value stream and not only CI. Deployment automation, environment provisioning, progressive delivery, observability and rollback are now first-class capabilities.

From containers to cloud native platforms: Kubernetes, Helm, GitOps and container-native ecosystems form a common operating model for distributed applications and increasingly for AI workloads.

Monitoring observability: Logs, metrics and traces are converging around open telemetry standards, with OpenTelemetry becoming a strategic interoperability layer.

AI augmented DevOps: AI assistants are expanding from code generation into test creation, code review, incident triage, root-cause analysis, remediation recommendations and operational forecasting.

From toolchain to engineering platform: Tool sprawl is being addressed through platform teams, standardised workflows, developer portals, golden paths, and common engineering metrics.

From DevOps metrics to outcome metrics: DORA delivery metrics are increasingly being combined with reliability, developer experience, security, cost and business outcome measures.